

Huzaifa Abdul Rehman

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SUMMARY

Computer Science undergraduate at FAST NUCES (6th semester) with strong focus on full-stack development and ML-driven systems. Experienced in building scalable web applications using Next.js, React, TypeScript, and Python, with solid foundations in database design, algorithms, and applied machine learning.

EDUCATION

FAST – National University of Computer and Emerging Sciences

BS Computer Science, CGPA: 3.56 (Dean's List – 4 semesters)

Karachi, Pakistan

Expected Aug 2027

Adamjee Government Science College

Intermediate (Pre-Engineering), Percentage: 87.36%

Karachi, Pakistan

PROJECTS

Electrolux Energy Management System (EMS) — Next.js, TypeScript, MySQL

Fall 2025

- Designed normalized MySQL database (16 tables, BCNF) using pre-seeded datasets to validate billing, metering, and payment workflows
- Built role-based SaaS portal with secure authentication (NextAuth) enabling interactions between administrators, employees, and customers
- Developed real-time analytics dashboards with consumption charts and complete end-to-end workflows for billing and payment tracking

FAST Academic Hub — React, Vite, Tailwind CSS, PWA

Fall 2025

- Built a Progressive Web App to unify university timetable planning, attendance tracking, and GPA calculation, transforming complex Excel-based schedules into a clean, searchable, offline-capable platform
- Implemented features including smart search filters, automatic course conflict detection, personal timetable generation with color-coded visualization, and GPA/CGPA tracking across multiple semesters
- Optimized for offline use with Workbox service workers, zero-backend architecture, mobile-first responsive UI, and deployed on Vercel, improving usability and performance for FAST students

Web Crawler & Search Engine — Python, NLP, TF-IDF

Fall 2025

- Built BFS-based web crawler compliant with robots.txt, achieving 90%+ word extraction accuracy processing 15-50 pages per crawl
- Implemented TF-IDF search engine with sub-second query response for 100 documents with 1000+ indexed terms
- Developed NLP pipeline using Porter Stemmer, stopword filtering, and cosine similarity ranking

ML-Enhanced Dijkstra Routing System — Python, scikit-learn, XGBoost, NetworkX, OSMnx

Fall 2025

- Achieved 90.4% R^2 accuracy with Gradient Boosting model, reducing average route travel time by 18.87% (100.3 seconds saved per route)
- Engineered 15+ features including road types, congestion factors, and capacity scores to improve model performance
- Built production-ready routing system with 162-215ms execution per route and interactive map visualization of 10,011 nodes

ACTIVITIES & INVOLVEMENT

Volunteer – Non-Profit Organization

Karachi, Pakistan

Community Operations & Inventory Management

- Managed inventory tracking and distribution of donated products
- Coordinated logistics for iftar distribution campaigns and beach cleanup drives

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C++

Frameworks & Libraries: React, Next.js, Node.js, Express.js, Tailwind CSS

ML & Data: scikit-learn, XGBoost, TF-IDF, NLP, Feature Engineering

Databases: MySQL, PostgreSQL, MongoDB

Tools & Platforms: Git, GitHub, AWS, Vercel, Postman, Jupyter Notebook